

The Metric Role from Continuation Comparison

Proto-Metric Representation as a Secondary Stability Structure

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May 2026

Abstract

The preceding reconstruction sequence established non-vacuous realized persistence, bounded standing-localization, and pre-topological continuation-neighborhood domains. This paper derives the next physical role: metricity at the level of stable comparison. It asks what is required for local continuation domains to be compared stably under overlap, restoration, and standing-preserving transport. The answer is not a primitive metric: primitive distance, coordinate ordering, graph weights, or metric tensors would select comparison before the relevant continuation roles had been fixed.

The main result is a carrier-induction theorem. Given induced continuation-neighborhood structure, stable overlap witnesses, path-coherent standing transport, and quotient-stable redescription, admissible comparison witnesses determine a quotient comparison carrier over local continuation domains. The associated representation is proto-metric because it preserves identity, distinguishability, overlap coherence, and comparative continuation burden where admissible witnesses exist.

The physical-role claim is that the metric role first enters as quotient-stable comparative representation, before numerical distance, topology, manifolds, Lorentzian signature, curvature, field support, or spacetime dynamics are assumed. Standard metric spaces, Riemannian or Lorentzian metrics, geodesic structure, and gravitational dynamics are downstream specializations of this comparison-carrier role.

1 Introduction

1.1 Theorem claim and physical role

The paper derives the metric role before its standard mathematical specialization. The theorem does not import a metric. It derives the comparison carrier that any later metric structure must preserve:

$$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \wedge \text{OverlapStab}_D \wedge \text{TransportCoherence}_D \wedge \text{QuotStab}_D \Rightarrow \text{ProtoMetricRep}_D.$$

The physical-role claim is that metricity first appears as stable comparative representation among local continuation domains. Real-valued distance, metric tensors, Lorentzian signature, geodesics, curvature, and gravitational equations are later strengthening conditions on this derived comparison role.

The first three papers in the realized-physics arc establish a staged reconstruction ladder. Minimal non-vacuity gives the persistence floor: a target typed as carrying realized persistence inside a fixed admissible continuation regime cannot remain null with respect to its own persistence-sensitive reconstruction structure [11]. In schematic form,

$$\text{RealPers}_D \Rightarrow \text{NonVacRec}_D.$$

Standing localization then shows that non-vacuous standing-bearing persistence cannot remain merely global, diffuse, trace-only, hidden, or unbounded once perturbation and restorative continuation constraints are admitted [13]:

$$\text{NonVacRec}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv_B} \neq \emptyset.$$

Continuation-neighborhood induction then shows that bounded standing-localization classes cannot support cross-locus continuation without boundary-compatible adjacency, standing-verdict separability, and local continuation domains [10]:

$$[B_D]_{\equiv_B} \wedge \text{BoundaryFix}_D \wedge \text{ContCompat}_D \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{Sep}_D, \text{LCD}_D).$$

The present paper begins at this point. Local continuation domains are available, but stable comparison is not. Adjacency is continuation-relevant boundary compatibility, not distance. A continuation-neighborhood is not an open set. A local continuation domain is not a spatial region. Nothing in the preceding stage supplies length, magnitude, angle, curvature, metric balls, geodesics, coordinate charts, or a metric tensor.

The question is therefore narrow:

Once continuation-neighborhood domains exist, what additional structure is required for their stable comparison under overlap, restoration, and standing-preserving transport?

The answer is not a primitive metric. A primitive metric would select comparison, ordering, and separation before the required continuation roles have been fixed. Instead, comparative representation becomes admissible only as a secondary stability structure: it records stable comparative burdens among already-induced local continuation domains.

The distinction is essential:

Derived here	Deferred to later strengthening layers
comparative continuation burden	metric geometry
stable overlap representation	numerical distance
path-coherent transport comparison	geodesic or length structure
quotient-stable comparison carrier	metric tensor
comparative representability	topology, manifolds, and spacetime geometry

1.2 Main thesis

The controlling thesis is:

The first metric-facing structure is not primitive distance, geometry, or spacetime structure, but a quotient-stable representation of comparison witnesses among already-induced continuation-neighborhood domains.

In compact form, the main implication is

$$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \wedge \text{OverlapStab}_D \wedge \text{TransportCoherence}_D \wedge \text{QuotStab}_D \Rightarrow \text{ProtoMetricRep}_D.$$

The conclusion is deliberately weak compared with full metric geometry. ProtoMetricRep_D denotes a stable quotient comparison representation over local continuation domains. It does not denote a metric space in the standard mathematical sense, a Riemannian or Lorentzian metric, a coordinate distance, a graph distance, a causal interval, a field support, or a spacetime line element.

1.3 Scope

The paper establishes five claims.

- (i) **Primitive comparison import is inadmissible.** A metric, ordering, weighting, or comparison carrier cannot be introduced before localization and continuation-neighborhood structure without acting as a selector, representative chooser, ordering authority, or hidden comparison standard.
- (ii) **Adjacency is not comparative representation.** Continuation-relevant adjacency gives boundary-compatible relevance, not magnitude, length, distance, angle, curvature, or metric topology.
- (iii) **Stable overlap requires comparative invariance.** If local continuation domains overlap in standing-relevant structure, the overlap must remain invariant under lawful redescription, restoration, and same-domain transport.
- (iv) **Standing-preserving transport requires path-stable comparison.** Transport across admissible adjacency paths must preserve quotient identity, boundary traces, restoration status, and standing verdicts without arbitrary representative choice.
- (v) **Comparative representation is secondary.** A comparative carrier becomes admissible only after local domains, overlap witnesses, path coherence, and quotient stability are already fixed.

The paper does not yet specialize the comparison carrier into a standard metric space, topology, manifold, metric tensor, Lorentzian signature, finite propagation speed, causal cone, curvature, Einstein dynamics, gauge structure, particle support, or vacuum structure. Those require additional strengthening conditions. The positive result is the metric role itself at its earliest admissible layer: quotient-stable comparison among local continuation domains.

1.4 Relation to standard metric frameworks

Metric geometry normally begins with a set equipped with a distance function satisfying specified axioms [1, 2]. Topology studies neighborhood and open-set structure directly [15]. Riemannian and Lorentzian geometry introduce differentiable manifolds, tangent structure, metric tensors, and signature conditions [6, 16, 5]. Algebraic quantum field theory typically begins with spacetime regions and assigns local algebras to them [4, 3]. These frameworks enter after some locality, region, neighborhood, or distance structure is already available.

The present analysis targets an earlier layer. It asks why a comparison role should arise at all after pre-topological continuation-neighborhoods have been induced. The answer is that stable overlap and standing-preserving transport require comparative invariants. The induced comparison representation is the role-stable record of those invariants, but it is not mere notation: by concrete-domain and anchor-tensor-skin discipline, it must carry standing-bearing role structure rather than diagrammatic or coordinate skin [9, 8].

2 Metric strengthening hierarchy

The carrier induction theorem proves comparative representability at the comparison-carrier layer, not full metric geometry. The distinction is best treated as a hierarchy.

Level	Structure
M0	Continuation-neighborhood adjacency among bounded standing-localization classes.
M1	Stable overlap comparison among local continuation domains.
M2	Comparative continuation-burden representation in an internal admissible comparison carrier.
M3	Pseudometric-like representation after additional collapse-handling and composition conditions.
M4	Metric representation with identity of indiscernibles, symmetry when applicable, and triangle-type constraints.
M5	Manifold-compatible metric field with differentiable structure and local charts.
M6	Physical spacetime metric, for example Lorentzian metric structure with causal, dynamical, and empirical interpretation.

The present paper establishes the structural pressure toward M1-M2 and gives conditional access to M3 only when additional quotient and path-composition constraints are imposed. It does not establish M4-M6. In particular, no claim is made here about Riemannian distance, Lorentzian interval, metric compatibility, curvature, geodesic structure, or field equations.

3 Dependency order and type discipline

The argument is conditional on the earlier layers. It does not re-prove them.

Layer	Imported role
Minimal non-vacuity	Realized persistence requires non-vacuous reconstruction structure.
Standing localization	Standing-bearing persistence factors through bounded localization classes.
Continuation-neighborhood induction	Bounded loci induce boundary-compatible adjacency, separability, and local continuation domains under lawful continuation.
Boundary-trace fixation	Cross-domain comparison must be anchored by jointly reconstructible boundary traces.
AMetric boundary	No primitive metric, comparison scale, selector, coordinate ordering, or hidden comparison standard may be imported.
Scale-transport discipline	Transport must preserve standing-bearing quotient behavior under redescription and regime transition.
Concrete-domain discipline	Comparative representation cannot remain formal-only, trace-only, or skin-only.

Definition 3.1 (Fixed admissible realized persistence regime). A fixed admissible realized persistence regime D is a same-domain continuation setting in which standing-bearing persistence is evaluated without cross-domain laundering, hidden selector import, fail-open relabeling, or identity reset. All necessity claims in this paper are conditional on this fixed-domain setting.

Definition 3.2 (Local continuation domain). A local continuation domain is imported from the continuation-neighborhood layer. For a bounded locus L ,

$$\text{LCD}_D(L) = \{K : L \rightsquigarrow_{\text{Adj}_D}^* K\},$$

where $L \rightsquigarrow_{\text{Adj}_D}^* K$ denotes an admissible adjacency path preserving boundary compatibility, quotient behavior, target identity, restoration compatibility, and fail-closed standing status. The object $\text{LCD}_D(L)$ is not a spatial region. It is the continuation domain generated by admitted adjacency.

Definition 3.3 (Overlap witness). Let U, V be local continuation domains. An overlap witness $\text{OW}_D(U, V)$ is a shared continuation trace, boundary role, or quotient-preserving interface through which standing-bearing consequence is jointly reconstructible in both U and V . An overlap witness is not an observational record, coordinate intersection, set-theoretic overlap by notation, or diagrammatic contact.

Definition 3.4 (Stable overlap). $\text{OverlapStab}_D(U, V)$ holds when overlap witnesses between U and V remain jointly reconstructible under lawful redescription, restorative continuation, and admissible same-domain transport. Stable overlap excludes notation-only overlap, hidden representative selection, and overlap that changes standing verdicts under admissible redescription.

Definition 3.5 (Transport coherence). $\text{TransportCoherence}_D(U, V)$ holds when standing-bearing content can be carried across admissible adjacency paths from U to V while preserving quotient identity, boundary traces, restoration status, standing verdicts, and local-domain role. Transport coherence excludes identity reset, path-arbitrary relabeling, hidden selector import, coordinate drift, and domain drift.

Definition 3.6 (Quotient stability). QuotStab_D holds when standing-bearing quotient behavior is invariant under the comparisons and transports admitted in D . In particular, two descriptions that represent the same standing-bearing continuation role must remain equivalent under admissible redescription, while distinct standing-verdict classes must not collapse merely because a different representative, coordinate label, or path notation is chosen.

Definition 3.7 (Comparison witness). For local continuation domains U, V , let $\text{CompWit}_D(U, V)$ denote the class of admissible overlap, restoration, and transport witnesses that compare the standing-bearing role of U and V . A comparison witness may be a shared overlap trace, a quotient-preserving interface, a path-coherent transport role, or a restoration-burden role. It is admissible only when it is fixed by boundary-compatible continuation inside D . It is not a coordinate label, diagram edge, graph weight, external scalar, or observer-selected ordering.

Definition 3.8 (Witness equivalence). Two comparison witnesses $w, w' \in \text{CompWit}_D(U, V)$ are equivalent, written $w \sim_D w'$, iff they preserve the same standing-bearing quotient role under lawful same-domain redescription, restoration, and path-coherent transport. Witnesses are not equivalent merely because they are assigned the same notation; equivalence is fixed by preservation of boundary traces, quotient behavior, standing verdict, and local-domain role.

Definition 3.9 (Comparative carrier). The comparative carrier is the quotient carrier of admissible comparison witnesses:

$$C_D := \bigsqcup_{(U,V) \in \text{Dom}(\rho_D)} \text{CompWit}_D(U, V) / \sim_D,$$

where

$$\text{Dom}(\rho_D) := \{(U, V) : \text{CompWit}_D(U, V) \neq \emptyset\}.$$

Thus C_D is not chosen as a numerical scale. It is the collection of stable witness classes available inside the fixed regime. It may be ordered, partially ordered, preordered, directed, or otherwise role-stratified when those comparison relations are themselves witnessed. It need not be numerical, scalar-valued, linearly ordered, globally comparable, $\mathbb{R}_{\geq 0}$ -valued, total, symmetric, additive, Archimedean, continuous, or manifold-compatible.

Definition 3.10 (Quotient comparison representation). A quotient comparison representation, denoted by ProtoMetricRep_D at the proto-metric layer, is the partial map

$$\rho_D : \text{Dom}(\rho_D) \subseteq \text{LCD}_D \times \text{LCD}_D \longrightarrow C_D$$

defined by

$$\rho_D(U, V) = [w]_{\sim_D} \quad \text{for } w \in \text{CompWit}_D(U, V),$$

whenever the admissible witness class is unique up to $\sim_D$ and fixed by admissible overlap, transport, restoration, and quotient behavior. If multiple non-equivalent witnesses are admitted between the same ordered pair, the representation is relational rather than single-valued until additional regime conditions select a unique comparison role. The representation is partial because forcing comparison between domains with no admitted witness would import a hidden global selector. It represents stable comparative burden subject to the following minimal conditions:

- (P1) self-comparison is neutral or identity-fixed where the self-witness exists;
- (P2) non-collapsed separability classes remain distinguishable by admitted witness classes;
- (P3) comparison classes are stable under lawful redescription;
- (P4) admissible path composition preserves standing-bearing quotient behavior;
- (P5) no primitive metric, topology, coordinate system, manifold, graph metric, or hidden selector is imported.

This is weaker than a metric. It is the minimal quotient-stable comparative representation induced by stability requirements on already-formed continuation-neighborhood domains.

Remark 3.11 (Why the carrier may be weaker than a metric). A standard metric normally requires a real-valued distance satisfying non-negativity, identity of indiscernibles, symmetry, and the triangle inequality. None of these is assumed here. Prior to stronger geometric conditions, continuation and restoration burden may be partially ordered, direction-sensitive, path-sensitive, or regime-indexed. The first admissible object is therefore a quotient comparison carrier at the comparison-carrier layer, not a full metric space.

4 Comparison carrier discipline

The comparison carrier C_D is the first place where metric-like comparison behavior can be represented, but it is not selected as a scale. It is constructed from admissible comparison witnesses modulo same-role redescription. At this stage it is neither a numerical line nor a coordinate interval. It is a quotient role carrier constrained by standing-bearing continuation. Its admissible

values are the classes $[w]_{\sim_D}$ that preserve overlap, transport, restoration, and quotient behavior under lawful redescription.

A comparison carrier therefore has three duties. First, it must preserve *self-fixity*: comparing a local continuation domain with itself cannot introduce an artificial burden. Second, it must preserve *non-collapse*: local domains that remain standing-verdict separable must not be identified merely because a chosen representation lacks enough resolution. Third, it must preserve *composition discipline*: if standing-bearing content is carried through a chain of admitted adjacency relations, the comparison class of the composite path must be fixed by admissible intermediate boundary traces, not by arbitrary path notation.

These duties do not yet imply the metric axioms. They imply only that comparison classes must be stable enough to be reused across admissible redescription. A carrier that changes class when a diagram is redrawn is skin. A carrier that changes class when an equivalent representative is chosen is selector-dependent. A carrier that cannot distinguish separable standing verdicts collapses the continuation-neighborhood structure. A carrier that cannot compare admissible composite paths fails transport coherence. Quotient comparison representation is the surviving case: it is neither primitive metric import nor mere notation, but stable comparative role.

Remark 4.1 (Partiality and direction). The first comparison representation may be partial or directed, because ρ_D is defined only on admitted witness-bearing pairs. If restoration from U to V and restoration from V to U carry different admissibility burdens, forcing symmetry would erase standing-relevant directionality. If two domains have no admitted overlap or adjacency path, forcing a comparison value may import a hidden global selector. For this reason, partiality and directionality are allowed at the comparison-carrier layer. Symmetry and total comparability are later strengthening conditions, not initial requirements.

5 Primitive comparison import is inadmissible

Neither a metric nor a general comparison scale can be placed at the foundation of this layer without doing work that earlier papers explicitly refuse to import. A primitive metric would select which loci are close, which paths are shorter, which representatives compare, and which separation relations count before boundary-compatible continuation has fixed those roles. A primitive comparison scale would do the same work more abstractly by assigning order, weight, or burden before overlap and transport constraints had licensed such assignment. In the AASC setting either move is an AMetric boundary violation: a selection is being made without a same-domain admissibility authority [7].

Theorem 5.1 (No primitive comparison import). *Let D be a fixed admissible realized persistence regime prior to quotient comparison representation. A metric assignment, comparison carrier, ordering, weighting, or burden scale introduced before bounded localization, continuation-neighborhood structure, stable overlap, and transport coherence is inadmissible as a source of standing-bearing comparison unless its values are fixed by overlap witnesses, path-coherent transport, and quotient-stable redescription internal to D .*

Proof. Suppose a metric or comparison assignment m is introduced before the relevant localization and continuation-neighborhood roles are fixed. Then m must determine at least some comparative facts among loci or candidate domains: which pairs are near or comparable, which paths are short or burdensome, which separations matter, or which representatives supply the comparison target. Since these comparative facts precede boundary-compatible continuation, they cannot be

grounded by overlap witnesses, transport coherence, or quotient stability. Thus m functions as one of the following: an external selector, a representative chooser, a hidden ordering authority, a coordinate convention carrying standing force, or a default comparison standard. Each alternative violates fixed-domain admissibility. If m carries no standing-bearing role, it is skin-only notation and cannot ground quotient comparison representation. Therefore primitive comparison import, including primitive metric import, is inadmissible for standing-bearing comparison. $\square$

The theorem is not a denial that standard physics uses metrics. It is a priority claim. A metric may enter once its comparison role is fixed by already-induced continuation structure. It may not enter as the origin of that structure.

6 Adjacency is not comparative representation

The preceding paper derives continuation-relevant adjacency. It does not derive metric distance. This distinction is essential. Adjacency records that boundary-compatible standing-relevant consequence can pass, compare, constrain, or restore across loci. A metric records comparative magnitude, distance, separation, or burden. One may eventually represent adjacency by a graph and graph distance, or by metric neighborhoods, but that is a representation added after the adjacency role is fixed.

Lemma 6.1 (Adjacency is not comparative representation). *($\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D$) does not by itself determine a quotient comparison representation.*

Proof. Adjacency specifies continuation relevance between loci. It does not specify comparative magnitude among adjacent relations, relative burden of alternative paths, redescription-invariant overlap weights, or identity-stable comparison values. Two continuation-neighborhood structures can have the same adjacency relation but different transport burdens; conversely, a family of comparison values can be assigned arbitrarily to an adjacency relation without preserving standing-bearing quotient behavior. Therefore adjacency alone is insufficient for quotient comparison representation. $\square$

This lemma prevents a common overclaim. Local continuation domains provide the domain on which comparison may become meaningful. They do not supply the quotient comparison representation by themselves.

7 Stable overlap and comparative invariance

Stable comparative representation becomes necessary only when local continuation domains are not merely adjacent but stably comparable. The first source of comparison is overlap. If two domains share a standing-relevant boundary trace or quotient-preserving interface, the shared role must be recognized consistently across admissible redescription and restorative continuation.

Proposition 7.1 (Stable overlap requires comparative invariance). *If $\text{OverlapStab}_D(U, V)$ holds for local continuation domains U, V , then D must preserve a comparative invariant for the overlap role of U and V .*

Proof. Stable overlap means that the shared continuation trace or quotient-preserving interface remains jointly reconstructible in both domains under lawful redescription and restoration. If no comparative invariant is preserved, then the overlap can change role when descriptions,

representatives, or restoration paths change. The alleged overlap would be notation-only, path-dependent, or selector-dependent. In each case the overlap ceases to be stable in the standing-bearing sense. Therefore stable overlap requires an invariant comparative role. $\square$

The invariant need not be numerical. It may be an equivalence class, partial order, ordinal burden, restoration class, transport class, or other role carrier. The important point is not numeric magnitude but stable comparison under admissible transformation.

8 Transport coherence and path-stable comparison

The second source of comparison pressure is transport. If standing-bearing consequence moves through an admissible adjacency path, the result cannot depend on arbitrary path notation or representative choice. Otherwise the transported verdict is not a continuation invariant.

Proposition 8.1 (Transport requires path-stable comparison). *If $\text{TransportCoherence}_D(U, V)$ holds across admissible adjacency paths, then the comparison of transport burdens between U and V must be stable under admissible path redescription and quotient-preserving composition.*

Proof. Transport coherence requires that standing-bearing content carried from U to V preserve quotient identity, boundary traces, restoration status, standing verdicts, and local-domain role. Suppose transport burden is not path-stably comparable. Then two admissible redescriptions of the same standing-bearing transport can yield different comparison verdicts without a difference in boundary traces or quotient behavior. The difference is then produced by representative choice, path notation, hidden selector import, or domain drift. Each option is inadmissible. Hence transport coherence requires path-stable comparison. $\square$

This is the scale-transport lesson in a pre-metric setting. Transport may vary when a lawful regime transition allows variation; it may not vary by arbitrary same-domain representative choice [12, 14].

9 Comparative continuation burden

The comparative invariant induced by stable overlap and transport is best treated first as continuation burden, not as distance. It is not spatial distance, coordinate length, graph distance, or the value of a metric tensor. It is a stable comparison class of admissible continuation between local continuation domains.

Definition 9.1 (Continuation burden). For local continuation domains U, V , the continuation burden $\delta_D(U, V)$ is the standing-relevant comparative load required to preserve overlap, restoration status, quotient identity, and transport coherence between U and V . At this layer $\delta_D(U, V)$ denotes a witness class, not necessarily a numerical quantity. It need not be scalar, additive, symmetric, total, or linearly ordered. It may be represented in C_D by $\rho_D(U, V) = [w]_{\sim_D}$ only when $(U, V) \in \text{Dom}(\rho_D)$.

Criterion 9.2 (Minimal continuation-burden representation conditions). A continuation-burden representation may serve as admissible quotient comparison only if it satisfies the following minimal conditions.

(D1) $\delta_D(U, U)$ is neutral or self-fixed.

- (D2) If U and V are non-collapsed standing-verdict separability classes, $\delta_D(U, V)$ does not collapse them by representation alone.
- (D3) $\delta_D(U, V)$ is invariant under lawful redescription preserving boundary traces and quotient behavior.
- (D4) Admissible path composition preserves the comparison class of transported standing-bearing content.
- (D5) Any asymmetry, partiality, or path-dependence must be grounded by admitted continuation structure, not by arbitrary notation.

Symmetry is not imposed at this stage. Directed restoration or transport burdens can occur before a stronger reversible or metric-symmetric layer has been established. The triangle inequality is also not imposed. It is a later strengthening condition associated with path composition and metric representability, not with the first appearance of stable comparison. For this reason continuation burden is not treated as distance except when downstream strengthening conditions are explicitly under discussion.

10 Path composition without premature metric axioms

Path composition is the main pressure toward stable comparative representation, but it is not yet the triangle inequality. An admissible adjacency path

$$U = L_0 \text{Adj}_D L_1 \text{Adj}_D \cdots \text{Adj}_D L_n = V$$

contains intermediate boundary traces. Each intermediate trace must preserve quotient identity and restoration status. A proposed comparison assignment must therefore remain coherent when the same standing-bearing transport role is decomposed into finer or coarser admissible path descriptions.

This coherence requirement has a comparison-theoretic role because it controls how comparison behaves under composition. Nevertheless, it is weaker than metric additivity or subadditivity. Before a numerical carrier is justified, the most that can be required is preservation of comparison class under admissible path redescription. When boundary traces are composable, write

$$[w_1]_{\sim_D} \odot_D [w_2]_{\sim_D}$$

for the partial composition of comparison classes. The operation $\odot_D$ is defined only when the intermediate continuation boundary preserves quotient identity and standing-bearing role. If a composite path is admissibly equivalent to a direct transport role, their comparison classes must agree. If they differ, the difference must be grounded in a standing-bearing intermediate burden, not in arbitrary decomposition.

Proposition 10.1 (Composition discipline for quotient comparison). *Let $p : U \rightsquigarrow_{\text{Adj}_D}^* V$ and $q : U \rightsquigarrow_{\text{Adj}_D}^* V$ be admissible adjacency paths representing the same standing-bearing transport role under quotient-stable redescription. Let $\odot_D$ denote the partial composition of witness classes when intermediate boundary traces are composable. Any admissible quotient comparison representation must assign p and q the same comparison class in C_D , unless an admitted boundary witness distinguishes their standing-bearing burden.*

Proof. If p and q represent the same transport role and no boundary witness distinguishes them, assigning different comparison classes separates quotient-equivalent presentations. That violates quotient stability. If the different classes are justified by an admitted boundary witness, the paths no longer represent the same standing-bearing burden. When a path is decomposed through composable intermediate traces, $\odot_D$ records only admitted composition of witness classes. Therefore comparison status is invariant exactly up to admitted standing-bearing distinction. $\square$

This proposition is the pre-metric precursor of path-comparison consistency. It does not assert a numerical length functional. It asserts that comparison cannot depend on arbitrary redescription of the same transport role.

11 Carrier-induction theorem

We now state the main constructive result.

Theorem 11.1 (Carrier induction). *Let D be a fixed admissible realized persistence regime with induced continuation-neighborhood structure $(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D)$. If local continuation domains in D admit stable overlap witnesses, path-coherent standing transport, and quotient-stable comparison under admissible redescription, then a quotient comparison carrier supporting a weak proto-metric representation is induced as a secondary stability structure:*

$$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \wedge \text{OverlapStab}_D \wedge \text{TransportCoherence}_D \wedge \text{QuotStab}_D \Rightarrow \text{ProtoMetricRep}_D.$$

Proof. By the continuation-neighborhood result, local continuation domains are already available. Stable overlap supplies jointly reconstructible overlap witnesses between such domains. By the stable-overlap proposition, those witnesses require comparative invariants under lawful redescription and restoration. Transport coherence supplies admissible path relations through which standing-bearing content can be carried without identity reset, hidden selector import, or domain drift. By the transport proposition, such carrying requires path-stable comparison. Quotient stability ensures that equivalent standing-bearing roles remain equivalent under comparison while non-collapsed standing-verdict distinctions are not erased by representative change.

Construct $\text{CompWit}_D(U, V)$ as the admissible witnesses available between local continuation domains U, V , and define $w \sim_D w'$ when w and w' preserve the same standing-bearing quotient role under lawful same-domain redescription. The quotient

$$C_D := \bigsqcup_{(U,V) \in \text{Dom}(\rho_D)} \text{CompWit}_D(U, V) / \sim_D$$

is therefore fixed by the stable-overlap, transport, and quotient conditions rather than by an external scale; here $\text{Dom}(\rho_D)$ is the admitted witness-bearing domain of comparison. Define the partial representation by $\rho_D(U, V) = [w]_{\sim_D}$ for $w \in \text{CompWit}_D(U, V)$ exactly when the comparison role for (U, V) is fixed up to $\sim_D$. When several non-equivalent witnesses are admitted, the same construction yields a comparison relation rather than a single-valued map. In either form it preserves self-comparison where self-witnesses exist, distinguishes non-collapsed standing-verdict classes when admitted witnesses distinguish them, remains stable under lawful redescription, and respects admissible path-composition behavior through $\odot_D$. Therefore a quotient comparison carrier, and hence a weak proto-metric representation, is induced as a secondary stability structure. $\square$

Corollary 11.2 (Secondary status of comparative representation). *Comparative representation cannot be the source of the localization and continuation-neighborhood roles presupposed in the theorem. In this reconstruction sequence, comparative representation is secondary to bounded localization, boundary-compatible adjacency, stable overlap, and path-coherent transport.*

Proof. The theorem’s antecedent contains the structures required before the quotient comparison representation is available. A metric or comparison carrier introduced before them would violate the no primitive comparison import theorem. Hence comparative representation is secondary in this sequence. $\square$

12 Comparison carriers and quotient discipline

The conclusion is carried by a quotient comparison carrier, not necessarily by a numerical distance. This is important because the first admissible comparison-like structure must preserve the quotient behavior already fixed by the regime. If the carrier is too coarse, it erases separability. If it is too fine, it distinguishes representatives that the fixed domain treats as the same standing-bearing class. The carrier must therefore sit at the role level required by stable overlap and transport.

Proposition 12.1 (Carrier admissibility). *A comparative carrier C_D is admissible for quotient comparison representation only if it satisfies four constraints:*

- (C1) *it is generated by overlap witnesses or path-coherent transport classes;*
- (C2) *its equivalence classes are invariant under lawful same-domain redescription;*
- (C3) *it does not split standing-bearing quotient classes by representative choice;*
- (C4) *it does not identify standing-verdict separability classes that the regime must preserve.*

Proof. If the carrier is not generated by admitted witnesses, it imports comparison by fiat. If its classes change under lawful redescription, it is not a role-level invariant. If it distinguishes representatives inside a single standing-bearing quotient class, it mistakes notation for structure. If it identifies separability classes that the regime keeps distinct, it collapses local-domain structure. These are exactly the admissibility failures excluded by quotient stability. Hence any admissible carrier must satisfy the four constraints. $\square$

This proposition clarifies why the theorem does not simply declare a distance function. A distance function is one possible later representation of an admissible carrier. It is not the carrier itself at the first stabilization stage.

13 Conditional strengthening toward standard metrics

The transition from quotient comparison representation to standard metric geometry requires additional conditions. These conditions may be natural in later physical settings, but none is available for free at the present layer.

Proposition 13.1 (Conditional symmetry). *If admissible standing-bearing transport between local continuation domains is reversible with invariant burden under reversal, then the induced quotient comparison can be strengthened by a symmetry condition:*

$$\rho_D(U, V) \equiv \rho_D(V, U).$$

Absent reversible burden, symmetry is not licensed.

Proof. Symmetry requires that comparison from U to V and from V to U preserve the same standing-bearing quotient and burden class. If reversible transport with invariant burden is available, the two directions represent the same comparison role. If not, identifying them would erase direction-sensitive restoration or transport burden. Therefore symmetry is conditional on reversible same-domain transport. $\square$

Proposition 13.2 (Conditional triangle behavior). *If admissible path composition is burden-subadditive under quotient-stable concatenation, then a triangle-style constraint can be imposed on the induced comparison carrier. Without such composition discipline, triangle behavior is not forced by quotient comparison representation alone.*

Proof. A triangle inequality expresses a relation between direct comparison and composed comparison through an intermediate domain. Such a relation requires a lawful burden-composition operation and a subadditivity rule. The induced carrier supplies stable comparison classes, but it does not by itself supply an additive or subadditive composition law. Hence triangle-like behavior follows only when the additional composition rule is admitted. $\square$

Remark 13.3 (Conditional scalar representation). If the comparative carrier is total, Archimedean or otherwise scalar-representable, and stable under redescription, then a real-valued distance-like representation may be introduced. If the carrier is partial, directed, or regime-indexed, scalar representation is not generally available. This is a downstream representation condition, not part of the carrier induction theorem.

These conditional strengthening results are possible downstream extensions, not proto-axioms. Metric spaces, manifold-compatible metric fields, and Lorentzian metrics require additional symmetry, composition, scalarization, smoothness, and signature conditions beyond the quotient comparison carrier derived here.

14 Standard metric axioms as later strengthening conditions

The carrier induction theorem does not assume the standard metric axioms, but it explains where pressure toward them would come from. Each axiom becomes meaningful only after a corresponding continuation role has been fixed.

Metric-style condition	Continuation-theoretic pressure
Neutral self-distance	Self-comparison cannot introduce new standing burden.
Identity of indiscernibles	Non-identical local domains must remain distinguishable when standing-verdict separability requires it.
Non-negativity or ordered burden	Comparison values must not invert restoration or transport burden by representation alone.
Symmetry	Directional restoration burdens must collapse or be balanced by additional reversible-transport conditions.
Triangle-type constraint	Composite path comparison must be governed by admissible boundary witnesses rather than arbitrary decomposition.
Topology from metric balls	Continuation-neighborhoods must be strengthened into an open-set or convergence structure, which is not shown here.

This table is not a derivation of metric geometry. It is a dependency map. It shows why the familiar axioms are not primitive at the present layer. They are admissible only when the corresponding continuation roles have been stabilized strongly enough to support them.

15 Separability and nontrivial comparative distinguishability

Paper 3 derives standing-verdict separability: standing failure, perturbation burden, restoration obligation, or identity reset in one locus does not automatically determine the corresponding standing verdict elsewhere absent admitted adjacency paths. An admissible quotient comparison representation must preserve this non-collapse.

Proposition 15.1 (Separability requires stable distinguishability). *If local continuation domains U, V are standing-verdict separable in D , then any admissible quotient comparison representation over U, V must preserve their non-collapse as distinguishable comparison roles.*

Proof. Standing-verdict separability means that the standing status of one domain is not automatically determined by the other except through admitted adjacency paths. If a quotient comparison representation collapses U and V into the same comparison role without admitted quotient identification, it erases the separability distinction. If it distinguishes them by arbitrary label, it imports a selector. Therefore admissible representation must preserve their non-collapse through stable comparison roles grounded in overlap and transport structure. $\square$

This proposition explains why quotient comparison representation cannot be universally trivial. It must represent at least the distinctions required by standing-verdict separability.

16 Collapse modes for inadmissible comparison assignments

Several tempting alternatives fail. These failures are useful because they mark the exact role that quotient comparison representation must occupy.

Lemma 16.1 (Skin-only comparison collapse). *A comparison assignment that changes only coordinate labels, diagram placement, notation, visual layout, or presentation skin without carrying standing-bearing comparison role is inadmissible in D .*

Proof. A skin-only assignment does not affect overlap stability, transport coherence, quotient behavior, restoration status, or standing-verdict distinguishability. It is therefore inert with respect to the role quotient comparison representation is supposed to play. If it is treated as carrying standing force despite being skin-only, the force is imported by fiat. Either way it cannot ground admissible quotient comparison representation. $\square$

Lemma 16.2 (Trivial comparison collapse). *If a proposed metric assignment gives all local continuation domains the same zero or universal-equal comparative separation, then it cannot represent nontrivial standing-verdict separability or transport burden.*

Proof. Universal-zero or universal-equal comparison erases distinctions among local domains unless those domains have already been quotient-identified by admissible continuation. But Paper 3 supplies non-collapsed local continuation domains and standing-verdict separability. Erasing these distinctions by representation alone collapses the standing-verdict structure. Therefore a universally trivial comparison assignment is inadmissible whenever nontrivial separability or transport burden exists. $\square$

Lemma 16.3 (Coordinate comparison import failure). *A coordinate distance or graph weight assigned without boundary-compatible overlap witnesses and transport coherence is inadmissible as a source of quotient comparison representation.*

Proof. Coordinate distance or graph weight can represent an already-fixed comparison role, but neither can generate that role by itself. Without overlap witnesses and transport coherence, coordinate distance selects comparison targets, magnitudes, and ordering by external convention. That selection is not fixed by the standing-bearing continuation regime. Hence it is a selector import, not admissible quotient comparison representation. $\square$

Lemma 16.4 (Hidden metric reservoir exclusion). *A latent metric structure inaccessible to admissible standing-bearing continuation cannot ground quotient comparison representation in D .*

Proof. If the latent structure is inaccessible to admissible continuation, it cannot stabilize overlap, transport, quotient behavior, restoration, or standing-verdict distinction. It is realization-inert for the present role. If it becomes accessible, it must do so through admissible boundary-compatible continuation and is no longer a hidden reservoir. Thus hidden metric structure cannot rescue primitive metric import. $\square$

Default comparison. A rule declaring that one path, coordinate, representative, or magnitude is the default comparison standard without boundary-witnessed role fixation is an AMetric import. A default may become admissible only after the domain supplies a reason why that comparison class is invariant under lawful redescription and transport.

17 Why graph distance is not the result

Because local continuation domains can be represented by a finite graph in the witness model, it is tempting to treat graph distance as the quotient comparison representation derived here. That would be premature. A graph distance counts edges, or weights paths, after a graph and a weighting rule have already been supplied. The present result is prior to that rule. It asks which edge weights, path burdens, or comparison values are admissible in the first place.

An unweighted graph supplies adjacency, not stable comparative representation. A weighted graph supplies comparison-like information only if the weights are fixed by standing-bearing overlap and transport structure. If the weights are selected by diagram placement, coordinate convenience, or arbitrary numerical choice, they are metric imports. If all weights are equal by fiat, the graph may erase distinctions among different restoration burdens. If path weights disagree under quotient-equivalent redescription, transport coherence fails. Thus graph distance is a possible later representation of comparative structure, not the structure itself.

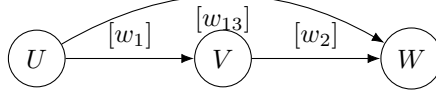
The same caution applies to manifold distance, causal interval, or algebraic localization radius. Each may become an excellent representation once the relevant downstream structure exists. None may be imported at the present layer to explain the appearance of stable comparison representation.

18 Finite witness model

A finite witness model clarifies the theorem without defining the underlying continuation-neighborhood structure. It instantiates one admissible realization of comparison transport rather than the general continuation-comparison structure itself. Let

$$G_D = (V, E, w)$$

where vertices V represent local continuation domains, edges E represent admitted boundary-compatible adjacency, and $w : E \rightarrow C_D$ assigns quotient comparison classes to admitted restoration or transport witnesses. Path comparison is interpreted only when intermediate boundary traces preserve standing-bearing quotient behavior and the partial composition $\odot_D$ of witness classes is defined. A quotient-consistent comparison class, not the drawing of the graph, carries the role. The graph is illustrative; it does not make graph structure primitive. It instantiates one admissible realization of comparison transport and does not define the general continuation-comparison structure.



stable comparison requires quotient-coherent agreement
between direct and composed admissible transports

Figure 1: A finite witness for quotient comparison carrier induction. Edge labels denote quotient comparison classes grounded by boundary-compatible overlap witnesses and transport coherence. The diagram illustrates comparison roles; it does not define the underlying continuation-neighborhood structure.

The quotient role is explicit. If the direct comparison class $[w_{13}]_{\sim_D}$ and the composed comparison class $[w_1]_{\sim_D} \odot_D [w_2]_{\sim_D}$ are both defined and represent the same standing-bearing transport under quotient-stable redescription, then

$$[w_{13}]_{\sim_D} = [w_1]_{\sim_D} \odot_D [w_2]_{\sim_D}.$$

If they differ, the difference must be witnessed by an admitted boundary role. Otherwise the comparison assignment has distinguished equivalent transport presentations by notation alone.

The finite witness model instantiates one admissible realization of comparison transport; it does not define the general continuation-comparison structure. It exhibits five cases.

- (a) **No comparison classes.** The graph may encode adjacency, but it lacks stable comparative burden. This is continuation-neighborhood structure without stable comparative representation.
- (b) **Arbitrary classes.** Assigning classes by coordinate or diagram choice imports a selector and violates the AMetric boundary.
- (c) **Universal-zero or universal-equal classes.** All standing-verdict separability and transport burden are erased unless the domains were already quotient-identified.
- (d) **Path-inconsistent classes.** If $[w_{13}]_{\sim_D}$ and $[w_1]_{\sim_D} \odot_D [w_2]_{\sim_D}$ disagree without an admitted quotient reason, transport coherence fails.
- (e) **Boundary-witnessed stable classes.** When comparison classes are fixed by stable overlap witnesses and quotient-coherent transport, they instantiate quotient comparison representation rather than selector weighting.

Thus the finite model instantiates the general theorem: quotient comparison representation is not an arbitrary weighting of adjacency. It is the stable comparative role left after selector imports, trivial collapse, and path incoherence are excluded.

19 Route exhaustion

The available routes can be expressed without enumerating every presentation-level variant.

Theorem 19.1 (Comparison route exhaustion). *Any candidate source of comparison representation at this layer falls into one of the following classes:*

- (i) *externally imposed comparison: primitive metrics, coordinates, graph weights, representative labels, or default orderings;*
- (ii) *internally unstable comparison: assignments that change under lawful redescription, restoration, or admissible path refinement;*
- (iii) *quotient-incoherent comparison: assignments that split equivalent standing-bearing roles or collapse standing-verdict distinctions the regime preserves;*
- (iv) *overlap-stabilized comparison: witness classes fixed by stable overlap, path-coherent transport, and quotient-stable redescription.*

The first three classes fail as admissible comparison representation for a fixed standing-bearing continuation regime. The fourth class is the surviving case and yields stable comparative representation, not full metric geometry.

Proof. Externally imposed comparison fails by the no primitive comparison import theorem: its values are not fixed by overlap witnesses, boundary traces, or path-coherent transport. Internally unstable comparison fails because a value that changes under lawful redescription or path refinement is not a standing-bearing comparison invariant. Quotient-incoherent comparison fails because it either distinguishes equivalent transport presentations by notation alone or identifies standing-verdict roles that the regime keeps distinct. The remaining class is comparison fixed by stable overlap witnesses, transport coherence, and quotient stability. By the carrier induction theorem, this class induces a quotient comparison carrier. Any additional candidate either supplies such stable boundary-witnessed comparison or collapses into one of the failed classes above. $\square$

20 Relation to metric geometry and spacetime metrics

A standard metric space (X, d) presupposes a set X and a distance function satisfying metric axioms. The present construction does not begin with such an object. It begins with local continuation domains generated by admissible adjacency paths and asks what comparative representation is required for stable overlap and transport. The resulting partial comparison map ρ_D may later be strengthened into a pseudometric, metric, length structure, or manifold-compatible metric only if additional axioms are justified.

The same caution applies to spacetime metrics. The result does not derive $g_{\mu\nu}$, Lorentzian signature, curvature, causal cones, geodesics, Einstein equations, or gravitational dynamics. It supplies a prior role constraint: any later spacetime metric must stabilize comparison among

local continuation domains under admissible overlap and transport. If a candidate metric cannot preserve those roles, it is not admissible as a realized metric structure in this arc.

This position is compatible with standard geometry as a downstream strengthening. Once a manifold, differentiable structure, and physical field role are established by later arguments, familiar metric notions may be recovered or constrained. The present paper denies that such structures may be assumed at the comparison-carrier layer.

21 Normal form

The result can be written as the following normal form:

$$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \wedge \text{OverlapStab}_D \wedge \text{TransportCoherence}_D \wedge \text{QuotStab}_D \Rightarrow \text{ProtoMetricRep}_D$$

The left-hand side is prior in the reconstruction order. Bounded standing-localization and continuation-neighborhood structure must already be available. Stable overlap and path-coherent transport then induce an admissible quotient comparison carrier. Quotient comparison representation is secondary because the comparison role it represents is fixed only after localization and continuation-neighborhood coherence have been established.

22 Conclusion

Paper 1 established that realized persistence cannot be reconstruction-null. Paper 2 established that standing-bearing persistence must localize through bounded continuation-sensitive loci. Paper 3 established that bounded loci induce pre-topological continuation-neighborhood structure under boundary-compatible continuation. The present analysis derives the next physical role: stable comparative representation, the role later occupied by metric structure.

No metric geometry, manifold structure, coordinate system, topology, tensor field, causal interval, propagation law, or spacetime metric is assumed. The result is intentionally prior to those forms: it establishes a quotient-stable comparison carrier at the comparison-carrier layer. Full metric spaces, manifold-compatible metric fields, Lorentzian spacetime metrics, geodesics, curvature, field support, and spacetime dynamics remain downstream.

The physical role of the result is that the paper does not merely avoid metric geometry. It derives the metric role before its mature formal species: stable comparison among local continuation domains under overlap, restoration, transport, quotient discipline, and boundary compatibility. Any later metric formalism in this arc must preserve that role or else import comparison by fiat.

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